

Laurent Colbois

APPLIED COMPUTER VISION SPECIALIST · PHD IN MACHINE LEARNING & BIOMETRICS

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"Applying AI expertise to create meaningful real-world impact through interdisciplinary collaboration."

Summary

Applied Computer Vision Specialist with a PhD from UNIL/Idiap, experienced in translating complex AI technologies into practical solutions for diverse stakeholders. Proven ability to bridge technical and non-technical audiences through outreach to forensic experts, law enforcement, and business stakeholders. Specialized in interdisciplinary collaboration, having worked across domains including biometrics, audio processing, and music generation. Seeking to apply AI expertise to meaningful real-world challenges with positive societal impact, particularly in environmental or scientific applications.

Work Experience

Idiap Research Institute

Martigny, Switzerland

POSTDOCTORAL RESEARCHER - APPLIED AI FOR FORENSIC SCIENCE

Apr. 2025 - Mar. 2026

- Led interdisciplinary project bringing together AI researchers, human cognition experts, software developers, and legal professionals to develop a forensic data collection platform
- Facilitated communication between technical and non-technical stakeholders, translating requirements across domains and ensuring alignment on project goals and constraints
- Developed explainable AI methods for face recognition to support forensic practitioners in understanding and trusting AI-assisted decision-making
- Demonstrated adaptability by rapidly pivoting to startup innovation during a 9-day hackathon, developing an AI music generation concept and successfully pitching to diverse audiences (technical experts, potential customers, investors)
- Applied rigorous evaluation methodologies to assess real-world applicability and trustworthiness of AI systems in high-stakes forensic contexts

Idiap Research Institute

Martigny, Switzerland

PHD RESEARCHER - AI APPLICATIONS IN FORENSIC SCIENCE

Jan. 2021 - Mar. 2025

- Conducted applied research on detecting AI-generated images in forensic contexts, directly addressing emerging challenges faced by border control and law enforcement agencies
- Delivered outreach presentations on AI and deepfakes to forensic science community and police forces, making technical concepts accessible to non-technical audiences
- Collaborated with forensic practitioners to understand real-world constraints and requirements, shaping research to address practical needs rather than purely academic questions
- Mentored four master's students from diverse backgrounds, adapting technical guidance to different learning styles and career goals
- Published research with strong emphasis on reproducibility and practical applicability, enabling practitioners to build on findings

Idiap Research Institute

Martigny, Switzerland

RESEARCH INTERN

Jul. - Dec. 2020

- Created synthetic datasets to enable more accessible and privacy-preserving benchmarking of face recognition systems
- Evaluated practical applicability of synthetic data for real-world face recognition applications

Logitech

Lausanne, Switzerland

RESEARCH INTERN - APPLIED AUDIO AI

Feb. - Aug. 2019

- Developed AI-powered automatic equalization system for loudspeakers, translating audio processing research into consumer product features
- Collaborated with product teams to understand user needs and technical constraints, adapting ML solutions to real-world hardware limitations
- Presented technical prototypes to cross-functional teams, explaining trade-offs and implementation strategies to stakeholders

Education

- Thesis: Face morphing attacks in the era of deepfakes: risks, detection & source attribution

EPFL

M.Sc. IN COMPUTATIONAL SCIENCE & ENGINEERING

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing

EPFL

B.Sc. IN PHYSICS

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing

1 Publications

2026	Conference Paper (Under Review) , Benchmarking the Explanatory Quality of Open-Weight Vision-Language Models in Face Recognition	Face and Gesture
2024	Conference Paper , Evaluating the effectiveness of attack-agnostic features for morphing attack detection	IJCB
2023	Conference Paper , Approximating Optimal Morphing Attacks using Template Inversion	IJCB
2022	Conference Paper , On the detection of morphing attacks generated by GANs	BIOSIG
2022	Conference Paper , Are GAN-based morphs threatening face recognition?	ICASSP
2021	Conference Paper , On the use of automatically generated synthetic image datasets for benchmarking face recognition	IJCB

Skills

Domains	Computer vision, Deep learning, Vision-Language Models, Biometrics, Audio processing, Reproducible research, Cross-functional communication
ML & Programming	Python, PyTorch, Huggingface Transformers, vLLM, Scikit-Learn, Pandas, NumPy, C++ (basic)
MLOps & DevOps	Git, Docker, Apptainer/Singularity, Conda, Pip, Pixi, Snakemake, Prefect, Weights & Biases, MLFlow, Hydra, Slurm, Unix/Linux
Languages	French (Mother tongue), English (Professional proficiency), German (B2)